

ELEMENTARY LINEAR ALGEBRA
FALL 2025 EXAM 2 OF 3

*“Mathematics is not about numbers, equations, or algorithms; it’s about understanding..”
William Paul Thurston*

You have worked hard this semester! Apply your mind to all you have learnt these last 8 weeks!
Give it your best.!!!

1) This is a 60 minute Exam. You may use the entire class period if you desire.

- a) Calculator allowed
- b) No notes allowed
- c) No smart devices/internet allowed

By signing below, I _____ acknowledge that I will uphold academic integrity and only use the resources allowed for each section.

Signature:_____

Date:_____

Name:_____

You may do your scratch work here.

1. a) (**10pts**) Find the basis of the null space of the matrix below.

$$A = \begin{bmatrix} 1 & -3 & 5 & 0 \\ 1 & -2 & -1 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & -18 & 9 \end{bmatrix}$$

- b) (**5pts**) Find the basis of the column/range space, ($\text{Col}A$) of the matrix A from part (a)

- c) (**5pts**) From (a) and (b), are the columns of A linearly independent? Provide simple justification.

- d) (**5pts**) From (a), (b), and (c), does A have an inverse? Provide simple justification.

You may do your scratch work here.

2. **(10pts)** Is the set of vectors, $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 7 \end{bmatrix} \right\}$ a basis?

(Show your work or give a clear explanation!)

3. **(30pts)** Given that $\mathcal{B} = \left\{ \begin{bmatrix} 0 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ 6 \end{bmatrix} \right\}$ and $\mathcal{D} = \left\{ \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \end{bmatrix} \right\}$ are bases of $\mathbb{R}^2$

i) (10pts) Find a transformation matrix that takes vectors from the the $\mathcal{B}$ -coordinate system to the $\mathcal{D}$ -coordinate system.

ii) (5pts) Find the coordinates of $\vec{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ in the $\mathcal{B}$ -coordinate system.

You may do your scratch work here.

iii) (5pts) Find the coordinates of $\vec{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ in the $\mathcal{B}$ -coordinate system.

iv) (15pts) Using your transformation matrix in (a), verify that your $[\vec{x}]_{\mathcal{B}}$ from part (b) is transformed onto your $[\vec{x}]_{\mathcal{D}}$ from part (c).

4. a) (**25pts**) Determine which of the following vectors $\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4$, and $\vec{v}_5$ are eigenvectors of the matrix B .

$$B = \begin{bmatrix} -2 & -4 & 2 \\ -2 & 1 & 2 \\ 4 & 2 & 5 \end{bmatrix}$$

If $\vec{v}_i$ is an eigenvector, simply state "YES", and give the corresponding eigenvalue. Otherwise, state "NO". Show your work!

i) $\vec{v}_1 = \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix}$

ii) $\vec{v}_2 = \begin{bmatrix} 0 \\ \sqrt{2} \\ 1 \end{bmatrix}$

You may do your scratch work here.

$$\text{iii) } \vec{v}_3 = \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}$$

$$\text{iv) } \vec{v}_4 = \begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix}$$

$$\text{v) } \vec{v}_5 = \begin{bmatrix} 1 \\ 6 \\ 16 \end{bmatrix}$$

- b) (**10pts**) Using your answers in (b), construct the matrices P and D such that the matrix B from part (a) is diagonalizable (that is, $B = PDP^{-1}$. (**Do not compute a P^{-1} !**)

Flip over to try the bonus problems!

These are bonus problems, so there is no partial credit! Only a clear well written solution/explanation will receive full point. Anything else receives a 0.

- (5pts) Find the eigenvalues of

$$A = \begin{bmatrix} -2 & -4 & 2 & 3 & \sqrt{5} \\ 0 & 1 & 5 & 3 & 0 \\ 0 & 0 & 5 & 1 & -2 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix}$$

- (5pts) The matrix, $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ has eigenvalues $\lambda_1 = 2$ and $\lambda_2 = -1$.
Find the eigenvectors correspond to λ_1 and λ_2 .

Great job for all your hard work!